CN Practical 3: Create and Test a Standard Cross Network Cable

Aim: To create and test a standard crossover network cable using a crimping tool and LAN cable tester.

Theory:

A crossover cable is an Ethernet cable in which the wiring arrangement at one end differs from the arrangement at the other end. Traditionally, it was used to connect similar Ethernet devices directly, such as two computers, without an intermediate switch. Modern networking devices often support Auto-MDI-X, which automatically adjusts the transmit and receive pairs, so a crossover cable may not be necessary in those cases.

The T568A and T568B standards specify the colour arrangement of the wires inside an RJ-45 connector. For a traditional crossover cable, one end uses the T568A standard and the other uses the T568B standard. The transmit and receive pairs are crossed to allow direct communication between compatible devices.

Materials required:

- UTP Ethernet cable
- Two RJ-45 connectors
- Crimping tool
- LAN cable tester
- Cable stripper

T568A colour sequence:

Pin	Wire colour
1	White-Green
2	Green
3	White-Orange
4	Blue
5	White-Blue
6	Orange
7	White-Brown
8	Brown

Procedure:

1. Cut the UTP cable to the required length.
2. Strip the outer insulation carefully from both ends.
3. Separate and straighten the eight internal wires.
4. Arrange the wires at the first end according to the T568A standard.
5. Trim the wires evenly and insert them into an RJ-45 connector.
6. Crimp the connector firmly using the crimping tool.
7. Arrange the wires at the second end according to the T568B standard.
8. Insert the wires into the second RJ-45 connector and crimp it.
9. Connect both ends to a LAN cable tester.

10. Verify the wiring pattern and continuity using the tester.

Observation:

The cable was prepared with T568A wiring at one end and T568B wiring at the other end. The cable tester was used to check continuity and the crossover pattern. Correct wiring ensures that the transmit and receive pairs are crossed as required by the traditional crossover design.

Conclusion:

A standard crossover Ethernet cable was created and tested successfully. The practical demonstrated the differences between T568A and T568B wiring standards and explained how crossover cables were traditionally used for direct communication between similar networking devices.